pfi

Python for Fixed Income

COURSE 4 · MACHINE LEARNING

Regression Assumptions and Log Spreads

WEEK 1 · DAY 4

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Three assumptions behind every p-value

Linearity

The model has the right shape: its misses follow no trend.

Constant variance

A tight bond and a wide bond are missed by about the same.

Normality

The misses look roughly like a bell curve.

Standard errors, t statistics, p-values, and intervals come from formulas that assume all three. The errors cannot be seen; the residuals, the misses left on the training rows, can. Every check today reads the residuals.

821 invented bonds, spreads made by the course's script. Why two checks fail is written into that script.

Residuals and fitted values

The bp model, training rows

17 rating dummies (reference A), duration, 9 sector dummies (reference Consumer): 27 features, statsmodels OLS on 615 training rows, target OAS in bp.

R-squared 0.6768 on the training rows

Residual = actual minus fitted, in bp

First training bond, rated A+: OAS 49 bp, fitted 34.2 bp, residual 14.8 bp. The mean residual is always zero with an intercept; the pattern is what the checks read.

Allowed names

fitted spread

residual: actual minus fitted

Never called

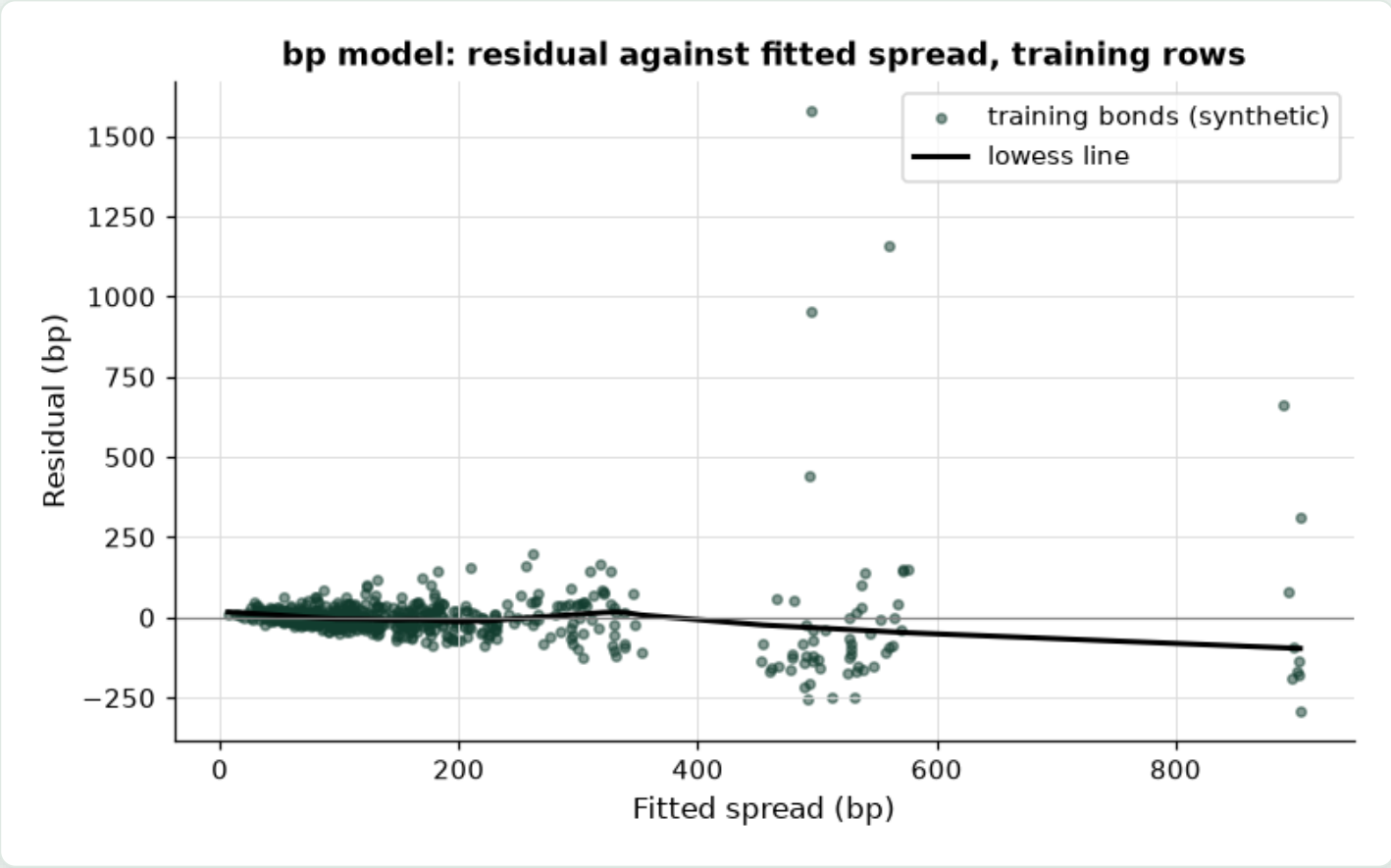
cheap, rich, mispriced

a signal, a verdict on a bond

How the linearity check works

The assumption	The average miss is zero wherever you look along the fitted line
The picture	Each training bond's residual in bp against its fitted spread
Lowess	At each point a small weighted line through the nearest points: a trend with no assumed shape
The reading	Flat at zero: the shape is right. A curve: too low in one part, too high in another
You control	The smoothing fraction, here 0.3 of the points

Linearity: residual against fitted



Fifth	Mean fitted	Mean residual
1	51.6 bp	6.2 bp
2	90.7 bp	-2.9 bp
3	127.7 bp	-2.0 bp
4	190.6 bp	-5.8 bp
5	442.8 bp	4.4 bp

Lowess: a small weighted line at each point, through its neighbours. Positive at both ends, negative between: the line bends.

A curve the model could have fitted.

How the constant variance check works

The assumption	The misses are about the same size at every fitted value
The fan	Residuals that widen as the fitted spread grows; standard errors become unreliable
By eye	Groups by fitted spread, with the residuals' spread in each group
Breusch-Pagan	Regress the squared residuals on the features; rows times R-squared, against chi-squared
What comes out	A statistic and a p-value: a small p-value rejects constant variance

Constant variance: the fan

Fifth	Residual sd
1	18.0 bp
2	23.7 bp
3	34.3 bp
4	43.5 bp
5	237.4 bp

Breusch-Pagan

Regress the squared residuals on the features. LM = rows x that R-squared, against chi-squared.

LM 51.63 on 27 df, p 0.00293

Reading a p-value

If the assumption held, how often would data show a pattern this strong by chance? Below 0.05 is called a rejection, by convention. It is not the probability the assumption is true.

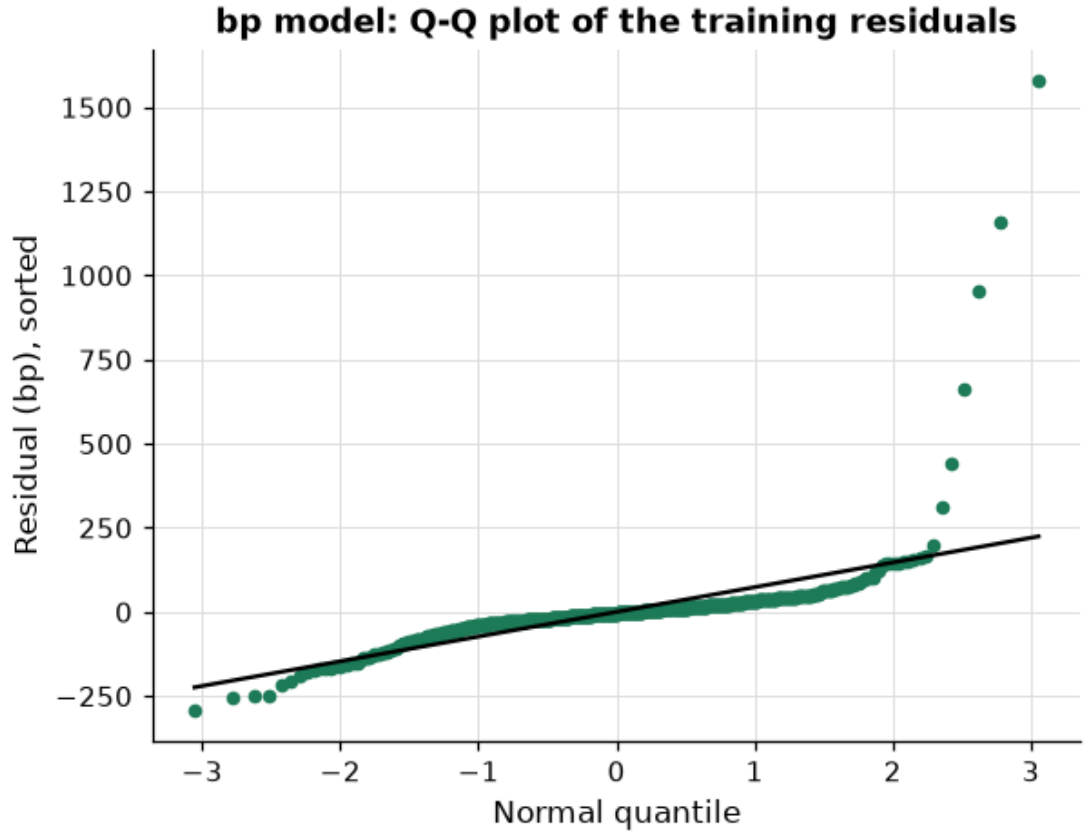
What a fan does: the coefficients stay a least squares fit; the printed standard errors, p-values, and intervals are not reliable.

Named beside it: Goldfeld-Quandt. Also named: standard errors that allow a fan (HC3).

How the normality check works

The assumption	The errors are roughly normal; p-values rest on it
Q-Q plot	Sorted residuals against where a normal would put them: a straight line if normal
The tail	A tail bending away from the line is heavier than normal
Two tests	Jarque-Bera: skewness and kurtosis. Shapiro-Wilk: sorted values against a normal's
What comes out	A picture and two p-values; a small p-value rejects normality

Normality: the Q-Q plot and two tests



8.20

skew (normal: 0)

100.23

excess kurtosis (normal: 0)

Test	Measures	p-value
Jarque-Bera	skew and kurtosis	< 0.001
Shapiro-Wilk	fit to normal quantiles	< 0.001

mlkit.residual_tests(residuals, exog)

Three p-values in one dictionary: Breusch-Pagan, Jarque-Bera, Shapiro-Wilk. Three new tests: 17 passed.

Why the bp model fans: the recipe multiplies

```
oas = base(score) x maturity factor x issuer effect x noise
```

Rating	Base (bp)	Noise sd (bp)
AAA	30	5.5
A	80	14.8
BBB	135	24.9
BB	260	48.0
B	440	81.1
CCC	900	166.0

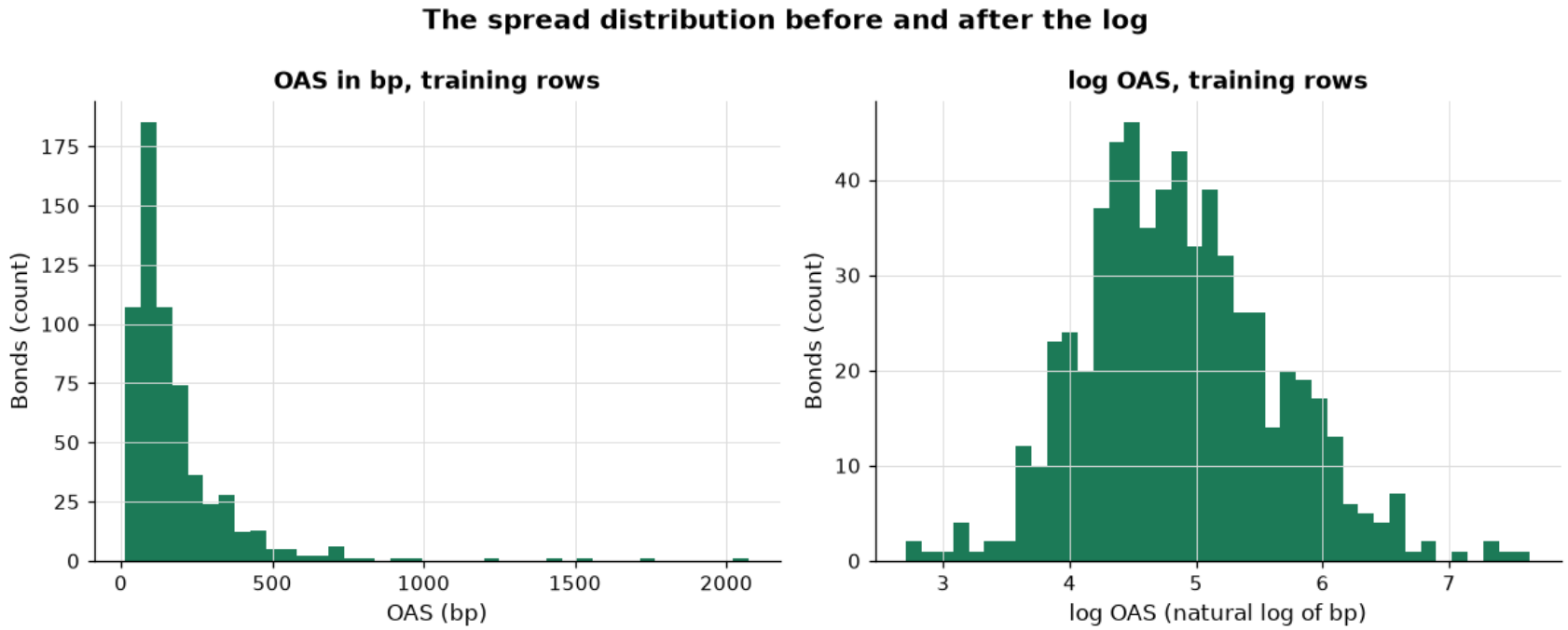
Noise sd = 18.4% of the spread

The same percentage, very different bp: that is the fan.

Logs turn multiplying into adding

log(oas) = log(base) + log(maturity factor) + log(issuer effect) + log(noise): the noise is 0.18 in logs for every bond.

The spreads before and after the log



4.43

skew of OAS in bp

0.38

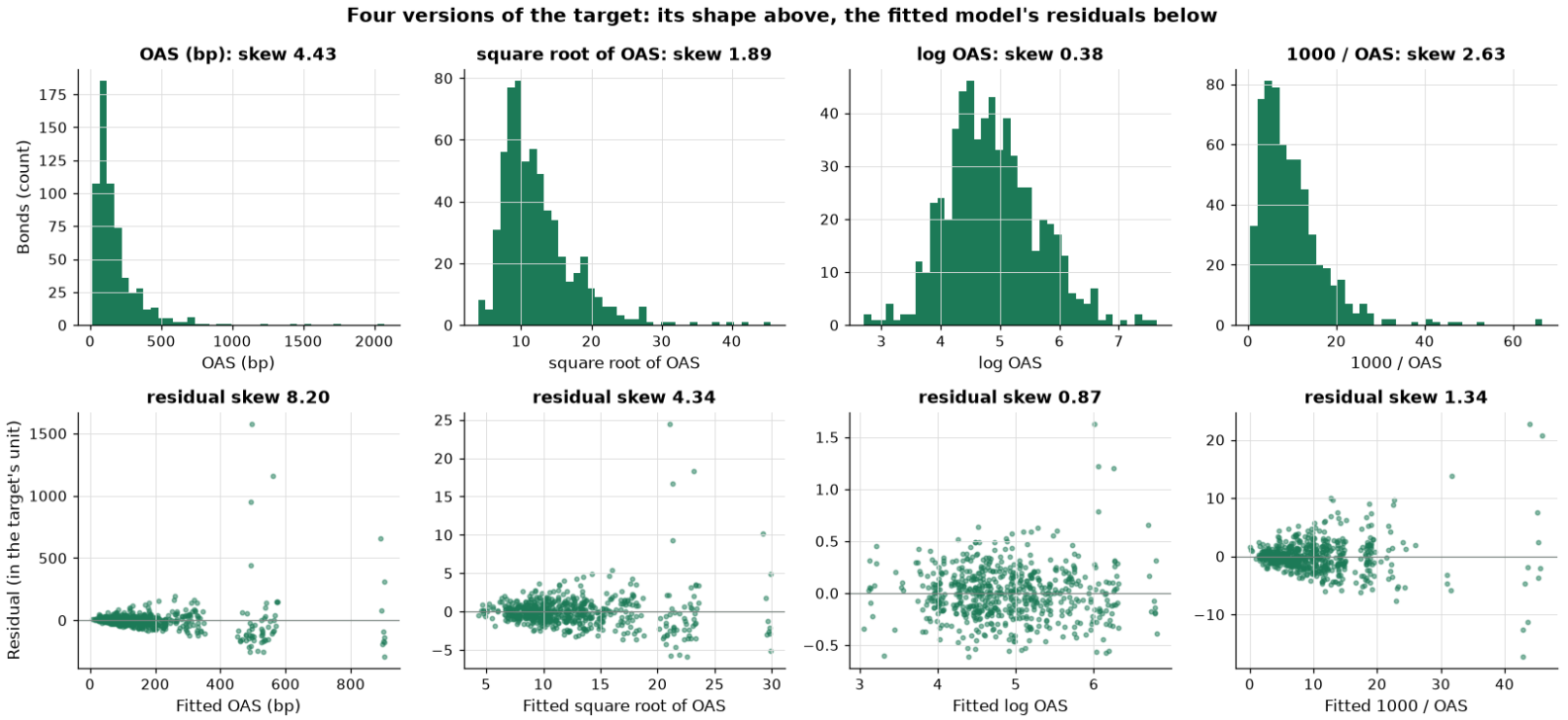
skew of log OAS (normal: 0)

The log pulls in wide spreads more than tight ones: the shape a recipe that multiplies needs.

How the transformation ladder works

The problem	A long right tail in the target breaks the assumptions
The ladder	Square root, then log, then reciprocal: each compresses large values more
The method	Fit the same model to each version of the target; compare the target and the residuals
What to read	Skew, the residual plot, and the Breusch-Pagan p-value for each rung
The choice	The mildest rung whose residuals stop fanning

Which transformation? A ladder

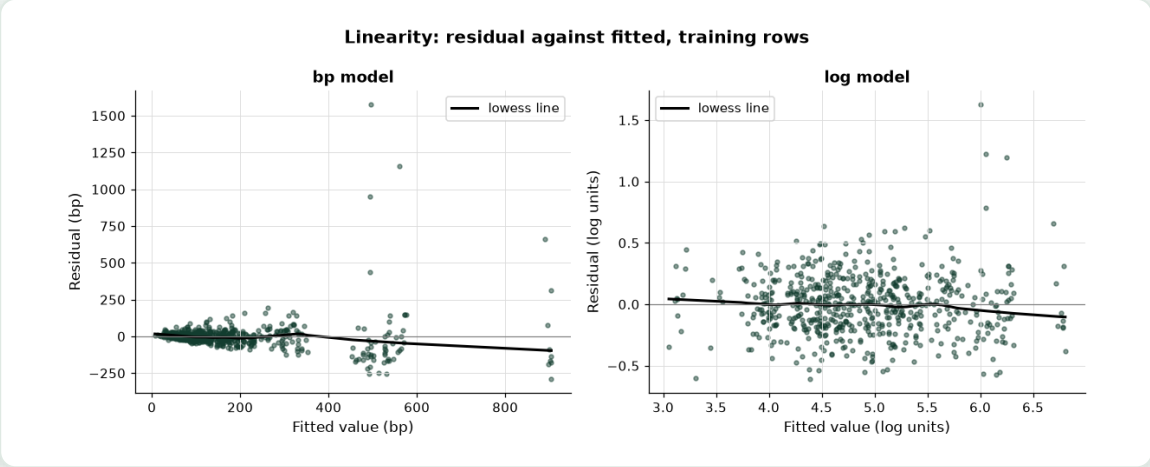


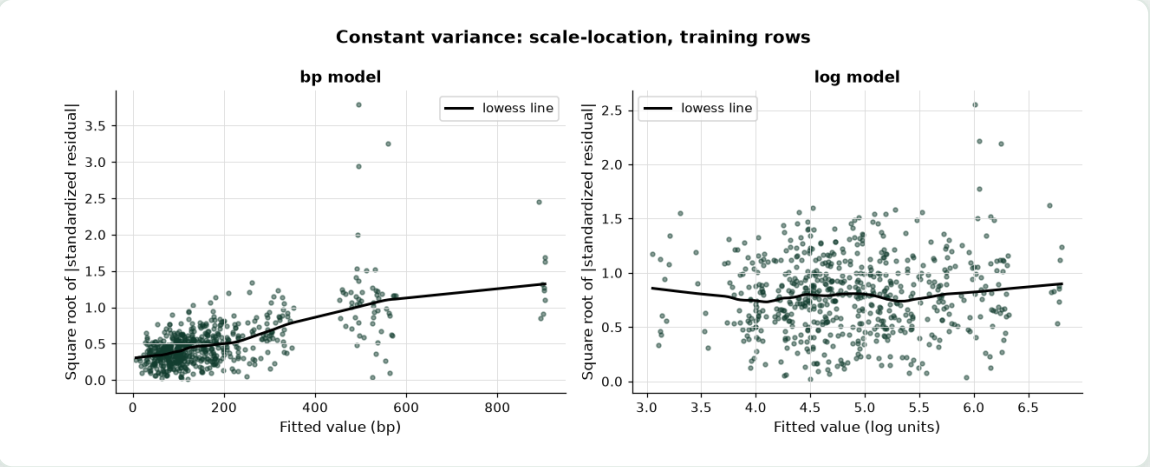
Skew of the target: 4.43 in bp, 1.89 square root, 0.38 log. The log is the first rung that is close to symmetric.

How a log model works

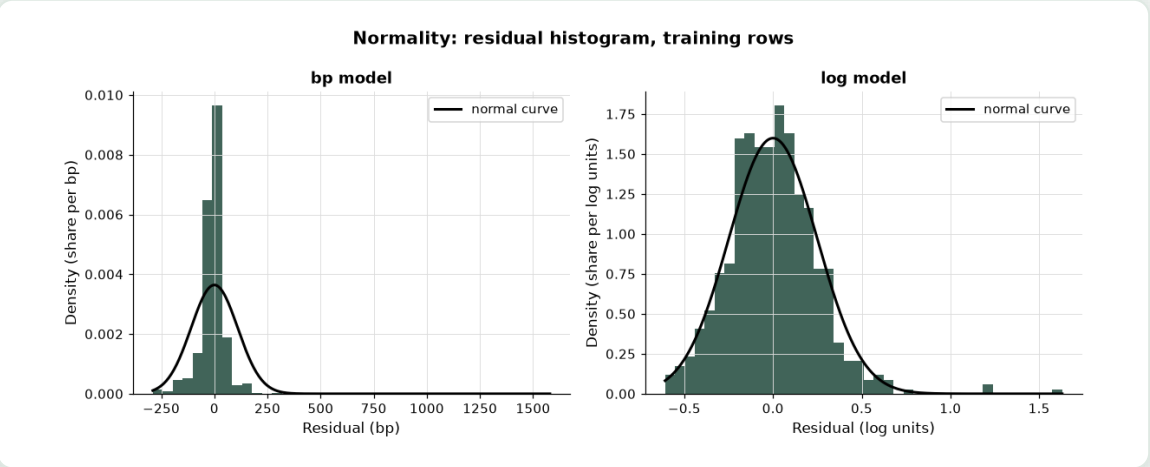
The target	The log of OAS instead of OAS, with the same features and training rows
Why	Each notch or year multiplies the spread by a factor, rather than adding bp
The fit	Least squares runs exactly as before, on log_oas
A coefficient	A change of about 100 times b percent per unit of the feature, for small b
What comes out	Log units throughout: convert fitted values back before reading them in bp

The four checks, before and after the log

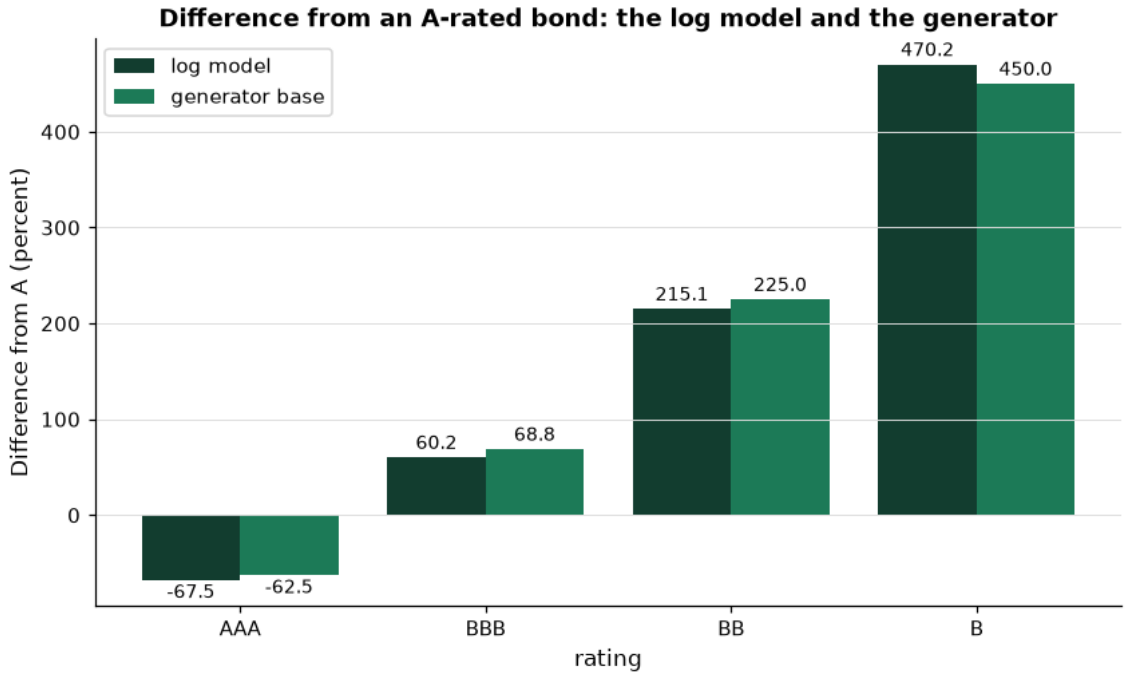








The log model: refit, recheck, read



Flat lowess line; residual sd 0.207 to 0.240 in four fifths, 0.326 in the top. Tests still reject (Breusch-Pagan 0.00786): a few points top right.

A coefficient b reads as

100 x (exp(b) - 1) percent

Duration 0.0200: 2.02 percent per year

vs A	Model	Generator
AAA	-67.5%	-62.5%
BBB	60.2%	68.8%
BB	215.1%	225.0%
B	470.2%	450.0%

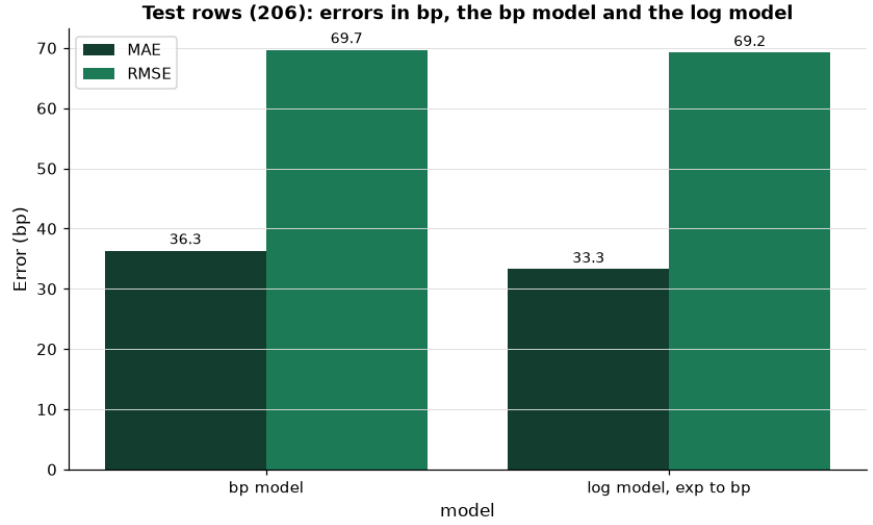
R-squared 0.8920 in logs: not comparable with 0.6768 in bp.

Back to bp: exp, the test MAE, and a median

206 test rows	MAE (bp)	RMSE (bp)	MAPE (%)	R-squared
bp model	36.3	69.7	21.2	0.8149
log model, exp to bp	33.3	69.2	18.5	0.8174

exp of a fitted log is a median-type value

108 of 206 test bonds sit above it. Mean actual 174.3 bp; mean of exp(fitted) 166.0 bp. No correction applied, and said so.

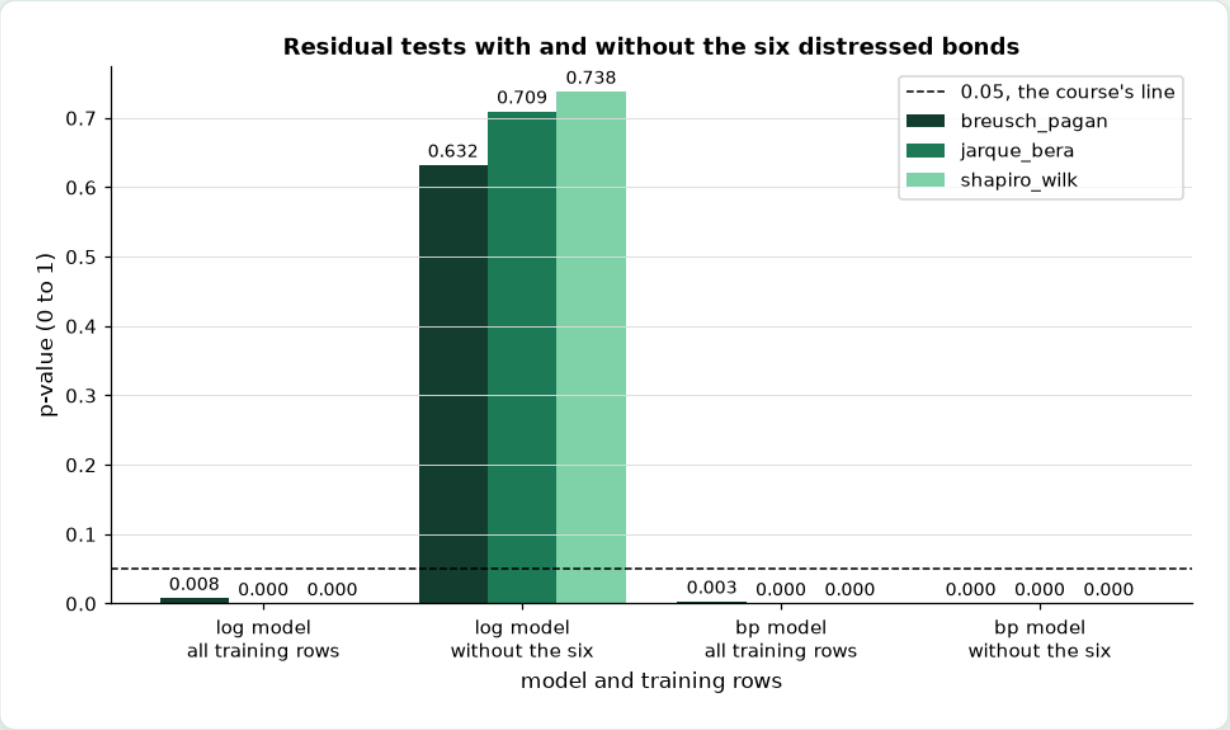


Main number, test MAE in bp: 33.3 for the log model against 36.3. Test rows used once.

The six distressed bonds

Rating	OAS (bp)	Gap (log)	Side
B	2074	1.74	training
B	1446	1.19	training
B-	1718	1.15	training
B	933	0.92	training
CCC+	1551	0.70	training
CCC+	1605	0.56	test

Each bond's log OAS minus its issuer's median, bonds rated B or lower. The next is at 0.43.

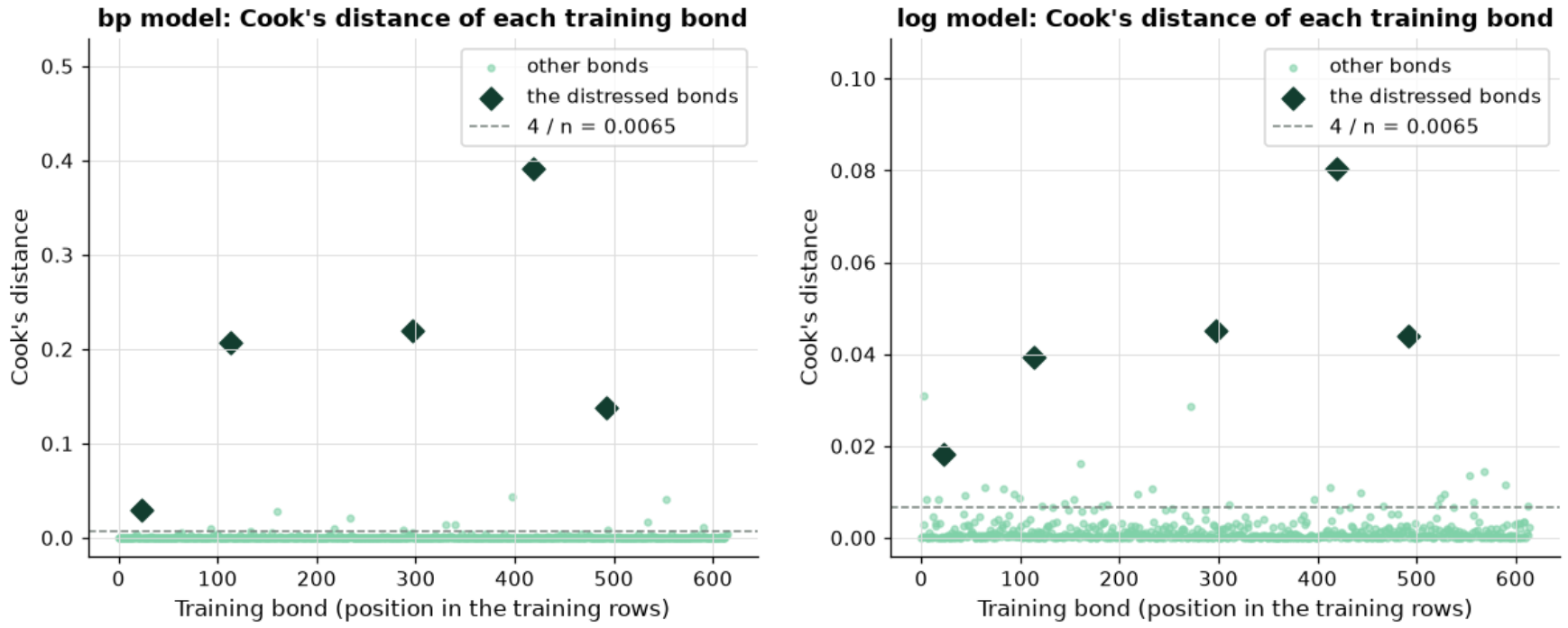


Kept: part of the data, not errors. Say which rows drive a test; never drop rows to pass one.

How Cook's distance works

The problem	A line can bend towards a few bonds far from the rest
The measure	How far all the fitted values would move if one bond were left out of the fit
When it is large	A bond far from the others in its features and also badly missed
You control	Which model to check, and what counts as large
What comes out	One number per training bond, drawn so the large ones stand out

Do a few bonds drive the fit? Cook's distance



Above the line: 18 bonds in bp (largest 0.392), 38 in logs (largest 0.080). A prompt to look, not a rule to delete.

Excel side by side

Excel side by side (tested in Excel)

`=LN(A2)  (B2 down)`	log OAS
`=TREND($A$2:$A$616,$C$2:$AC$616)  (AD2, spills)`	fitted, bp model
`=ROWS(AF2:AF616)*INDEX(LINEST(AF2:AF616,$C$2:$AC$616,TRUE,TRUE),3,1)`	LM 51.63
`=CHISQ.DIST.RT(AM1,COLUMNS($C$2:$AC$616))`	p 0.00293
`=NORM.S.INV((RANK.EQ(AI2,$AI$2:$AI$616,1)-0.5)/COUNT($AI$2:$AI$616))`	Q-Q point
`=AVERAGE(ABS(A2:A207-EXP(AD2:AD207)))`	33.3 bp

Q-Q positions (i - 0.5) / n are not probplot's: at the smallest residual -3.151 against -3.055. Without EXP the MAE formula returns 169.4: no unit.

Excel has no worksheet function for Breusch-Pagan, Jarque-Bera, or Shapiro-Wilk.

Save the day: git stash an experiment

1. Change

A square-root target: a third row in results.csv

2. Stash

git stash push -m: the committed file comes back

3. Compare

git stash show -p: the experiment as a diff

4. Drop

git stash drop: the log model stays

```
stash@{0}: On main: square-root target, an experiment
```

Square-root model: test R-squared 0.8238, MAE 34.0 bp against 33.3 bp; every residual check fails, even without the distressed bonds.

A stash is for an experiment you might want back within the hour; keep anything else in a commit.

The AI check

The docstring is the prompt. The code that comes back runs, and has one modelling bug.

1

Docstring

Which rows fit, which rows report, and the units: what you asked for.

2

Code back

Written for the course in the style of an AI answer. Read it.

3

Test first

Assert what the docstring promises. Run it: it fails.

4

Fix

Find the rule it broke, fix it, run the test again.

Today: the log model's test MAE is computed from fitted values left in logs and labelled bp: 169.4 against 33.3.

Code that runs is not the same as code that is right.

What you can do now

- Every p-value rests on three assumptions; check the residuals
- Lowess for shape, Breusch-Pagan for a fan, Q-Q and two tests for normality
- The bp model fans because the recipe multiplies
- log_oas: read b as 100 x (exp(b) - 1) percent; report in bp with exp
- Find the rows that drive a test, say so, keep them

Week 1, Day 5 Ridge, lasso, and pipelines, then a case study on log spreads